

Encoding



Encoding.m4a

Encoding is a technique that is used to convert categorical columns into numerical ones so that they can be fitted by machine learning models which only take numerical data. It is an important pre-processing step in a machine-learning project.

Example Of Encoding

Feature1	Feature2	Feature3	Target
4	Dhanmondi	6	B
3	Dhanmondi	2	A
6	Mirpur	4	B
6	Mirpur	3	A
7	Mirpur	4	B
2	Dhanmondi	7	B
3	Dhanmondi	2	A
5	Mirpur	4	B

For Feature2

Dhanmondi = 1
Mirpur = 2

For Target

A = 2
B = 3

Feature1	Feature2	Feature3	Target
4	1	6	3
3	1	2	2
6	2	4	3
6	2	3	2
7	2	4	3
2	1	7	3
3	1	2	2
5	2	4	3

Encoding

The most commonly used encoding techniques:

1. Label Encoding
2. One-Hot Encoding
3. Ordinal Encoding
4. Binary Encoding
5. Frequency Encoding

Label Encoding

Label encoding converts the categorical data into numerical ones, but it assigns a unique number(starting from 0) to each class of data.

Example Of Encoding

Feature1	Feature2	Feature3	Target
4	Dhanmondi	6	B
3	Dhanmondi	2	A
6	Mirpur	4	B
6	Mirpur	3	A
7	Mirpur	4	B
2	Dhanmondi	7	B
3	Dhanmondi	2	A
5	Mirpur	4	B

For Feature2

Dhanmondi = 0

Mirpur = 1

For Target

A = 0

B = 1

Feature1	Feature2	Feature3	Target
4	0	6	1
3	0	2	0
6	1	4	1
6	1	3	0
7	1	4	1
2	0	7	1
3	0	2	0
5	1	4	1

Label Encoding

Limitation of label Encoding

Label encoding converts the categorical data into numerical ones, but it assigns a unique number(starting from 0) to each class of data. This may lead to the generation of priority issues during model training of data sets. A label with a high value may be considered to have high priority than a label having a lower value.

Example For Limitation of Label Encoding

An attribute having output classes **Mexico**, **Paris**, and **Dubai**. On Label Encoding, this column lets **Mexico** is replaced with *0*, **Paris** is replaced with *1*, and **Dubai** is replaced with *2*.

With this, it can be interpreted that **Dubai** has high priority than **Mexico** and **Paris** while training the model, But actually, there is no such priority relation between these cities here.

One-Hot Encoding

One-Hot Encoding is a technique used in data preprocessing and machine learning to convert categorical data into a numerical format.

1. In One-Hot Encoding, each unique category or label in a categorical variable is represented by a binary vector.
2. The length of this binary vector is equal to the number of unique categories in the original categorical variable.

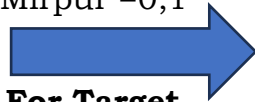
Example Of Encoding

Feature1	Feature2	Feature3	Target
4	Dhanmondi	6	B
3	Dhanmondi	2	A
6	Mirpur	4	B
6	Mirpur	3	A
7	Mirpur	4	B
2	Dhanmondi	7	B
3	Dhanmondi	2	A
5	Mirpur	4	B

For Feature2

Dhanmondi = 1,0

Mirpur =0,1



For Target

A = 1,0

B =0,1

Feature1	Feature2_Dhanmondi	Feature2_Mirpur	Feature3	Target_A	Target_B
4	1	0	6	0	1
3	1	0	2	1	0
6	0	1	4	0	1
6	0	1	3	1	0
7	0	1	4	0	1
2	1	0	7	0	1
3	1	0	2	1	0
5	0	1	4	0	1

One-Hot Encoding

Problem

F1	F2	F3	T
4	A	AA	1
3	B	BB	0
6	C	AA	1
6	D	CC	0
7	E	DD	1
2	F	AA	1
3	A	EE	0
5	G	FF	1

Number of features: 3

For F2

A = 1,0,0,0,0,0,0

B = 0,1,0,0,0,0,0

C = 0,0,1,0,0,0,0

D = 0,0,0,1,0,0,0

E = 0,0,0,0,1,0,0

F = 0,0,0,0,0,1,0

G = 0,0,0,0,0,0,1



For F3

AA = 1,0,0,0,0, 0

BB = 0,1,0,0,0, 0

CC = 0,0,1,0,0, 0

DD = 0,0,0,1,0, 0

EE = 0,0,0,0,1, 0

FF = 0,0,0,0,0, 1

F1	F2_A	F2_B	F2_C	F2_D	F2_E	F2_F	F2_G	F3_AA	F3_BB	F3_CC	F3_DD	F3_EE	F3_FF	T
4	1	0	0	0	0	0	0	1	0	0	0	0	0	1
3	0	1	0	0	0	0	0	0	1	0	0	0	0	0
6	0	0	1	0	0	0	0	1	0	0	0	0	0	1
6	0	0	0	1	0	0	0	0	0	1	0	0	0	0
7	0	0	0	0	1	0	0	0	0	0	1	0	0	1
2	0	0	0	0	0	1	0	1	0	0	0	0	0	1
3	1	0	0	0	0	0	0	0	0	0	0	1	0	0
5	0	0	0	0	0	0	1	0	0	0	0	0	1	1

Number of features: 14

One-Hot Encoding

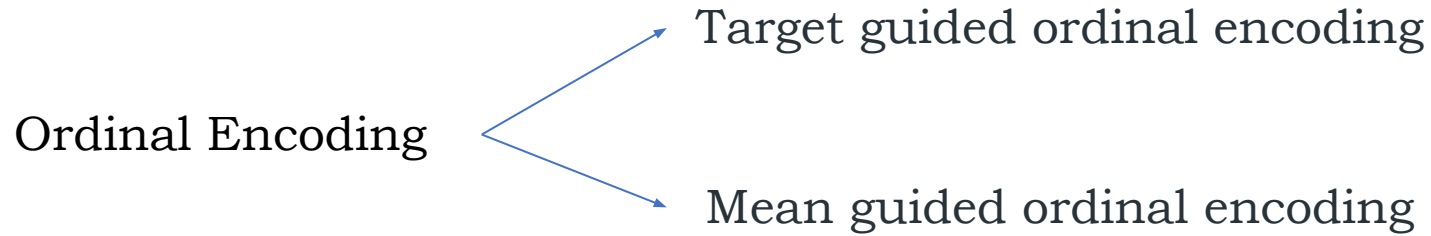
The advantages of using one hot encoding include:

It can help to avoid the problem of Label encoding. For example, if 00 represents “small”, 01 represents “medium”, and 10 represent “large”. There is no effect on the model accuracy as it does not contain any large value like Label-encoding.

The disadvantages of using one hot encoding include:

- 1.It can lead to increased dimensionality, as a separate column is created for each category in the variable. This can make the model more complex and slow to train.
- 2.It can lead to sparse data, as most observations will have a value of 0 in most of the one-hot encoded columns.
- 3.It can lead to overfitting, especially if there are many categories in the variable and the sample size is relatively small.

Ordinal Encoding



Employee Id	City	Highest Qualification	Salary
A100	Delhi	PhD	50000
A101	Delhi	BSc	30000
A102	Mumbai	MSc	45000
B101	Pune	BSc	25000
B102	Kolkata	PhD	48000
C100	Pune	MSc	30000
D103	Kolkata	MSc	44000

Apply **target guided ordinal encoding** on the **City** column

Step 1: sort the cities based on the corresponding salary. Now to do this we will take the mean of all the salaries of that particular city.

City	Mean
Delhi	40000
Mumbai	45000
Kolkata	46000
Pune	27500

Step 2: Based upon the mean of the salary the descending order of the city is :

Kolkata>Mumbai>Delhi>Pune

Ordinal Encoding

Step 3: Based on this order, we will rank the cities.

City	Rank
Kolkata	4
Mumbai	3
Delhi	2
Pune	1

Employee Id	City	Highest Qualification	Salary
A100	Delhi	PhD	50000
A101	Delhi	BSc	30000
A102	Mumbai	MSc	45000
B101	Pune	BSc	25000
B102	Kolkata	PhD	48000
C100	Pune	MSc	30000
D103	Kolkata	MSc	44000



Employee Id	City	Highest Qualification	Salary
A100	2	PhD	50000
A101	2	BSc	30000
A102	3	MSc	45000
B101	1	BSc	25000
B102	4	PhD	48000
C100	1	MSc	30000
D103	4	MSc	44000

Ordinal Encoding

Problem of target guided ordinal encoding

One problem with target guided ordinal encoding is that it can be sensitive to outliers. If there are a few data points with very high or very low salaries, then these data points can have a disproportionately large impact on the encoding of the categories. This can lead to the categories being encoded in a way that is not representative of the overall data distribution.

Overall, **target guided ordinal encoding is a powerful technique** for encoding categorical features in machine learning.

Ordinal Encoding

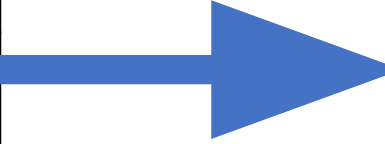
Apply **mean guided ordinal encoding** on the **Highest Qualification** column

Employee Id	City	Highest Qualification	Salary
A100	2	PhD	50000
A101	2	BSc	30000
A102	3	MSc	45000
B101	1	BSc	25000
B102	4	PhD	48000
C100	1	MSc	30000
D103	4	MSc	44000

Step 1: For each highest qualification we will find the mean of all the corresponding salary.

Highest Qualification	Mean Salary
PhD	49000
MSc	39666.67
Bsc	27500

Step 2 : Instead of ranking them based upon the mean value , we will encode this mean value corresponding to the respective highest qualification



Employee Id	City	Highest Qualification	Salary
A100	2	49000	50000
A101	2	27500	30000
A102	3	39666.67	45000
B101	1	27500	25000
B102	4	49000	48000
C100	1	39666.67	30000
D103	4	39666.67	44000

Ordinal Encoding

Problem of mean guided ordinal encoding

One problem with mean guided ordinal encoding is that it can be sensitive to outliers. If there are a few data points with very high or very low salaries, then these data points can have a disproportionately large impact on the encoding of the categories. This can lead to the categories being encoded in a way that is not representative of the overall data distribution.

Overall, **mean guided ordinal encoding is a powerful technique** for encoding categorical features in machine learning.

Binary Encoding

1. Binary encoding reduces the dimensionality by **representing categories as binary numbers**.
2. Each unique category is mapped to a unique binary code (sequence of binary digits).
3. The binary code is then split into individual binary digits, and each digit is represented by a separate feature column.

Suppose we have a categorical variable "Color" with three unique categories: "Red," "Green," and "Blue." Applying **One-hot encoding would create three new binary columns**, one for each color:

Color	Color_Red	Color_Green	Color_Blue
Red	1	0	0
Green	0	1	0
Blue	0	0	1

On the other hand, **applying binary encoding** would result in a different representation. Let's assume the following binary mappings for each color:

- ❖ Red: 00
- ❖ Green: 01
- ❖ Blue: 10

Then, using binary encoding, the dataset might look like this:

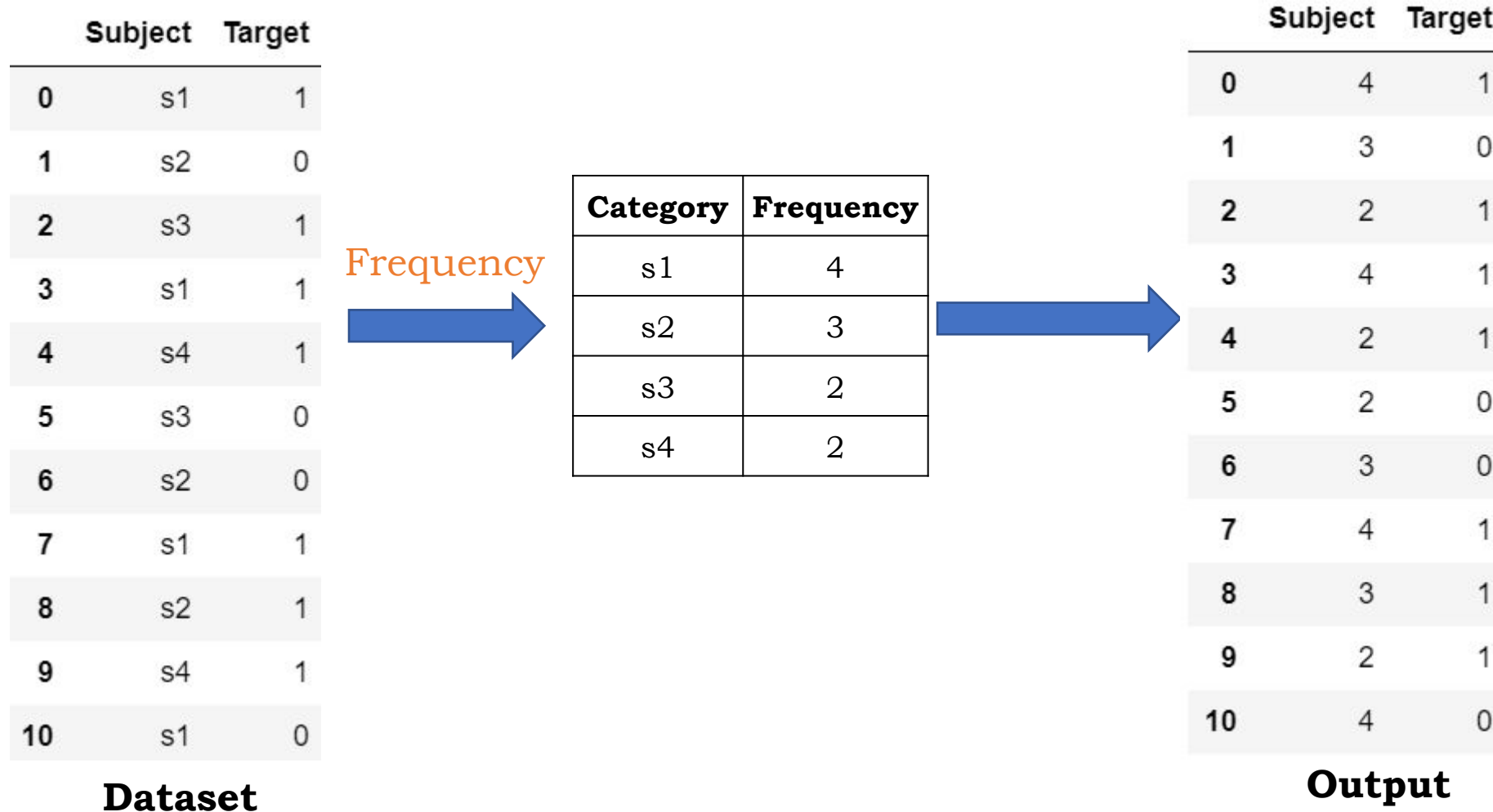
Color	Color_Bit_0	Color_Bit_1
Red	0	0
Green	0	1
Blue	1	0

Binary Encoding

Problem of Binary Encoding

Frequency Encoding

Frequency Encoding uses the frequency of the categories in data. In this method, we encode the categories with their frequency.



Frequency Encoding

Problem of Frequency Encoding

- 1. Equalizing Distinct Categories:** If two categories have similar frequencies, they will be encoded with similar values. This could cause the algorithm to treat them as more similar than they actually are, potentially leading to a loss of discrimination between categories.
- 2. Outliers and Rare Categories:** Categories with low frequencies might be sensitive to outliers. For instance, if an outlier occurs in a low-frequency category, it could disproportionately influence the encoded values, affecting the performance of the model.